

WCMS UNIT-1

2M:

1. What is content management system?

- A CMS is an application that is used to manage content, allowing multiple contributors to create, edit and publish.
- The definition of a CMS is an application (web- based), that provides capabilities for multiple users with different permission levels to manage (all or a section of) content, data or information of a website project, or intranet application.
- Managing content refers to creating, editing, archiving, publishing, collaborating on, reporting, distributing website content, data and information.

2. What are the common functions of CMS?

- **Content creation** allows users to easily create and format content
- **Content storage** stores content in one place, in a consistent fashion
- **Workflows** assigns permissions for managing content based on roles such as authors, editors and admins
- **Publishing** tells the software when and where the content should go live
- **Optimization** helps you improve digital experience and learn from your content

3. What are the types of content management system?

- Component Content Management System (CCMS)
- Document Management System (DMS)
- Enterprise Content Management System (ECM)
- Web Content Management System (WCMS)
- Digital Asset Management System (DAM)

4. Define web content management System?

- A Web Content Management System (WCMS) is a software system that enables the users to create, manage and deploy content on web pages.
- It is a type of content management system (CMS) that provides an organization with a way to manage digital information on a website through creating and maintaining content without prior knowledge of web programming or markup languages.

Or

- The Web Content Management System (WCMS) is a computer application used to create, edit, manage and publish content onto a website in an organized manner.

5. What are the types of WCMS?

- **Offline processing:** This type of WCMS processes content before publishing it to the live server.
- **Online processing:** Online processing systems employ templates on demand and whenever the user adds content to a webpage for publication.

- **Hybrid processing:** Hybrid processing systems use a combination of offline and online processing.

6. Define

a) CMA

=The content management application (CMA) is a user interface that enables users -- such as marketers and content creators -- to design, create, modify and remove content from the website without needing help from the IT department.

b) CDA

=The content delivery application (CDA) provides back-end services that take the content that users create in the CMA and turn it into the website that visitors can access.

7. Define DAM.

- With a digital asset management system, users can store, organize, and share digital content with ease.
- A DAM offers a simple, centralized library where clients, employees, or contractors can access digital content. These assets include audio, creative files, video, documents, and presentations.
- A DAM is cloud-based, so users can access content from anywhere.

8. What is the difference between WCMS and CMS.

WCMS	CMS
Specifically designed to manage web content.	Broader than WCMS, it can manage content across various platforms, not just web.
Primarily used for creating, managing, and deploying content on websites.	Can be used for managing documents, digital assets, records, and other content types across different environments (e.g., intranets, extranets, web).
Best suited for businesses or individuals looking to establish an online presence.	Suitable for organizations needing to manage a variety of content types and workflows, both online and offline.
Offers numerous plugins/extensions for enhancing website functionality.	May have modules/plugins for various content management and enterprise needs.

9. What is content and content format?

- **Content:** On the Internet, content is any information that is available for retrieval by the user, including web pages, images, music, audio, white papers, drivers and software downloads as well as training, educational and reference materials .

- **Content Format:** These are the ways you present your content to your audience, based on its medium, style, or layout. For example, you might have content formats such as text, video, audio, image, infographic, or carousel.

10. Mention any four Text file formats.

- .htm/.html/ files
- .xml files
- .doc files
- .odt files
- .pdf files

11. Mention any four common graphic file formats.

- .bmp files
- .cgm files
- .gif files
- .jpeg/.jpg files
- .JAS files
- PICT files
- .tiff files

12. Write various standards defined by Moving Picture Experts Group.

MPEG-1 - VHS-quality, commonly used for Video CD (VCD) and CD-ROM.

MPEG-2 - DVD-quality, commonly used for DVD, digital television, set-top boxes, etc.

MPEG-4 - Scalable delivery, used in various applications including the Internet, cell phones and television.

MPEG-4 is a patented collection of methods defining compression. It was used for compression of web-based audio/video streaming data and CD distribution, voice (telephone, videophone) and broadcast television applications. 36 Web Products and Services

13. What is digital graphics?

=Digital graphics are visual images or objects that are displayed on a digital device such as a computer.

They often combine both pictures and text to gain the attention of viewers on a website or digital space.

Or

Digital graphics combine art and technology to communicate ideas through images and the layout of web screens and printed pages. All digital images are either raster graphics or vector graphics.

14. What is raster Graphics?

=Raster graphics (also called a bitmap) is a way to construct a picture using small building blocks called pixels. When enough pixels are included close together, your eye stops seeing a collection of different points and starts seeing a whole image. Digital photographs are always rasterized.

15. What is vector graphics?

=Vector graphics use mathematical geometric elements (points, lines, curves, circles, and polygons) to create pictures. Because vector images are simply visual representations of geometric concepts (as opposed to raster files) they can be made infinitely large or small without losing image quality.

16. List advantages and disadvantages of raster graphics.

=Advantages of Raster Graphics

- High Detail: Raster graphics can display highly detailed images with subtle gradations of color, making them ideal for complex images like photographs.
- Realism: They can accurately represent real-world scenes and textures due to their ability to handle a wide range of colors and details.
- Rich Color Representation: Raster graphics can support millions of colors, allowing for vibrant and detailed images.
- Compatibility: Raster graphics formats like JPEG, PNG, GIF, and BMP are widely supported across different platforms and applications.

Disadvantages of Raster Graphics

- Large Files: High-resolution raster images can result in large file sizes, which can be problematic for storage and transmission.
- Loss of Quality: Scaling raster images up or down can lead to loss of quality and pixelation because they are resolution-dependent. Enlarging a raster image makes the pixels more visible, resulting in a blocky appearance.
- Complex Editing: Certain types of edits, such as changing the shape of an object within the image, can be more difficult with raster graphics compared to vector graphics.
- High Memory Usage: Large raster images can consume a significant amount of memory when being processed or edited, which can slow down performance on less powerful devices.

17. List advantages and disadvantages of vector graphics.

=Advantages of Vector Graphics

- Infinite Scalability: Vector graphics can be scaled up or down to any size without loss of quality. This makes them ideal for logos and other graphics that need to be resized frequently.
- Resolution Independence: They maintain their clarity and crispness at any resolution, making them perfect for both print and digital media.
- Smaller Files: Vector files are typically smaller than raster files because they only store the mathematical equations defining the shapes, not information about each individual pixel.
- Efficient Storage: The simplicity of the data in vector files allows for efficient storage and faster loading times.
- Easy Editing: Individual elements in vector graphics can be easily edited and manipulated without affecting the rest of the image. This makes them highly flexible for design changes.

- Precision: The mathematical nature of vector graphics allows for precise control over shapes, lines, and colors.

Disadvantages of Vector Graphics

- Less Detail: Vector graphics are not as effective at representing detailed and complex images, such as photographs, due to their reliance on shapes and lines rather than pixels.
- Complex Designs: Creating very detailed and realistic images can be challenging and time-consuming in vector format.
- Print Challenges: While vector graphics are excellent for large format printing, some print workflows and devices may prefer raster images, necessitating conversion which can sometimes introduce errors.
- Limited Format Compatibility: Although vector formats like SVG, AI, and EPS are common, they are not as universally supported as raster formats like JPEG or PNG.

18. Why do you need to connect content types and formats?

- First, it helps you create consistent and coherent content experiences for your audience, by ensuring that your content matches their expectations, needs, and preferences.
- Second, it helps you optimize your content for search engines and social media, by using the appropriate formats and metadata for each channel and platform.
- Third, it helps you manage your content more efficiently and effectively, by reducing duplication, simplifying updates, and enabling reuse and repurposing of your content.

19. List best practices for digital graphics.

- ☐ Limit the number of typefaces used.
- ☐ Go with a “less is more” approach.
- ☐ Play with scale.
- ☐ Reuse design elements.
- ☐ Mixing proportionate images and texts draws viewers’ eyes to specific areas of the design.

20. What is audio editing? Mention the use of audio editing.

=Audio editing is the process of manipulating sound recordings. This can be done for many reasons, such as to improve the quality of the recording, to remove unwanted noise or sounds, or to change the length or pitch of a particular audio clip.

Uses:

- Improve the quality of your audio content.
- Make your audio content more engaging.
- Save time and money.

21. Mention the types of audio editing.

22. What is cutting and fading?

23. What is fading and mixing?

=There are three main types of audio editing: cutting, fading, and mixing.

- **Cutting** is the most basic type of audio editing. To cut an audio file, select the portion of the file you want to remove and then delete it. This is often used to remove unwanted sections from a recording, such as pauses or mistakes.
- **Fading** is used to smooth out abrupt changes in volume. For example, if you have a recording of someone speaking and there is a sudden loud noise, you can use fading to gradually reduce the recording volume until the noise is gone.
- **Mixing** is used to combine multiple audio files into one. This is often used to create background music for a video or podcast. For example, you could mix several tracks to create an instrumental version of a song.

24. What are the benefits of audio editing?

- Improve the quality of your audio content.
- Make your audio content more engaging.
- Save time and money.

25. List and briefly explain any two audio editing tool.

- Audacity – it does almost everything you want, and it's open-source. It tends to be peoples 'go to' product.
- WavePad – WavePad is an audio editing software developed by NCH Software for both Windows and macOS.
- Apple Garage Band – GarageBand is a digital audio workstation developed by Apple, available for macOS and iOS.

LONG ANSWER QUESTIONS:

1. Explain the various types of content management systems. (6)

=TYPES OF CONTENT MANAGEMENT SYSTEMS:

1. Component Content Management System (CCMS)

A component content management system, or CCMS, differs from a standard CMS in that it organizes content at a granular level. Instead of managing content page by page, it takes words, phrases, paragraphs, or photos and stores them in a central repository.

2. Document Management System (DMS)

Paper is almost extinct. Tracking business files on paper is a thing of the past. A document management system (DMS) offers a paperless solution to manage, store, and track documents in a cloud. It provides an automated solution for uploading, processing, and sharing business documents without the hassle of printing, copying, or scanning.

3. Enterprise Content Management System (ECM)

An enterprise content management system collects, organizes, and delivers an organization's documentation, ensuring critical information is delivered to the correct audience (employees, customers, business stakeholders, etc.)

4. Web Content Management System (WCMS)

A WCMS lets users manage digital components of a website without prior knowledge of markup languages or web programming. A WCMS provides collaboration, authoring, and administration tools to help manage digital content. Unlike other CMSs, which deal with content destined for both the web and print, a WCMS handles exclusively web content.

5. Digital Asset Management System (DAM)

With a digital asset management system, users can store, organize, and share digital content with ease. A DAM offers a simple, centralized library where clients, employees, or contractors can access digital content. These assets include audio, creative files, video, documents, and presentations. A DAM is cloud-based, so users can access content from anywhere.

2. Explain the features of web content management systems. (6)

=The three key features of any WCMS are as follows:

- The ability to design, create and maintain personalized content for a website.
- The ability for editors to review and approve content prior to publication.
- An automated publishing process.

Additional features that a WCMS might include are:

Automated, standard templates. Provides users with templates that they can easily add new or existing content to, rather than designing their own templates.

Editing tools. Enables users to create and customize content, such as adding titles and headers to improve content structure, or designing webpages with drag-and-drop tools.

Easily installs plugins. Gives users the ability to extend a site's functionality with plugins or modules.

Software updates. Keeps the WCMS up to current web standards by updating software.

Workflow management. Ensures that authorized users review and approve content before publishing.

Collaboration tools. Enables multiple users to modify content.

Document management. Enables businesses to manage the document lifecycle, including creation, revisions, publication, archive and removal.

Multilingual. Enables the display of content in various languages.

Versioning. Enables editors to retrieve previous versions of content.

3. What are the Advantages of using WCMS? (4)

=A WCMS benefits businesses in the following ways:

Low cost. A WCMS is typically inexpensive, often being free or offering subscriptions that outweigh overall costs.

Easy to use. Most WCMS options are user-friendly for people who do not have a technical coding background to create or maintain content.

Easily customizable. A WCMS creates a universal layout that enables beginner users to create and customize front ends easily.

Workflow management. Administrators can control and personalize workflow management in a WCMS. Some WCMS options enable administrators to set up their own workflow management rules and provide them with a series of steps to set up each task.

Search engine optimization (SEO). A WCMS provides businesses with the editing tools required to improve search engine ranking. The WCMS should enable users to easily create content with the correct keywords, provide meta information -- such as description, keywords and alternative text for images -- and link content within the text.

4. What are the disadvantages of using WCMS? (4)

High cost for larger implementations. A WCMS for larger companies can be expensive because it may require extensive training and certifications. WCMS maintenance can also be expensive because the software may require upgrades and licensing updates.

Latency issues. Larger systems can become slower with time if the business does not keep the hardware up to date or if the cache files grow too large.

Security risks. If an administrator does not regularly patch the WCMS for security threats, it remains vulnerable to hackers. To minimize security risks, administrators must monitor and maintain the many moving parts of a WCMS -- such as the web server software, MySQL and any plugins or add-ons.

5. Explain the various popular content types. (4/6)

=Content types are the categories or classifications of your content, based on its purpose, function, or characteristics. For example, you might have content types such as blog posts, products, events, testimonials, or FAQs.

Most popular types of content:

The best websites create a mix of different content types to keep their readers interested. While there are hundreds of different types of web content some of the important content types are mentioned below:

Articles:

Articles are a type of web content that explore a specific idea or topic in detail. They often include interviews with experts, quotes from various sources, and data to back up the main points. Articles are usually longer than other types of online content, typically ranging from 1,000 to 3,000 words. This length allows for a thorough exploration of the subject.

Blog posts:

Blog posts are short articles typically between 500 and 750 words long. They cover a wide range of subjects, including industry news, profiles of important people, and reviews of new products. Blog posts are designed to be quick and easy to read, providing useful or interesting information in a concise format. They aim to engage readers with timely and relevant content without being too lengthy.

Surveys:

Surveys are a valuable addition to your content, as they gauge reader interest and boost engagement with your brand. They are interactive and fun, making them one of the most

engaging types of content for your audience. This helps create a stronger connection with your readers.

Quizzes:

Quizzes engage your readers by presenting a question on one page and the answer on the next. This setup increases the chances of ad revenue and click-throughs, making quizzes a fun and effective way to interact with your audience.

Photos

Images help make your content more visually appealing, and contribute to the overall user experience. Additionally, optimizing your photos with alt-text and captions to your images can even help with your rankings on Google Images.

Videos

Using videos is a great way to diversify your content and add visual interest to your pages. These can be videos you take yourself or videos from sources like YouTube or Vimeo. (Just make sure to give credit where credit is due.)

E-books:

E-books offer your readers more detailed information that they can download or read directly on your site. They enhance your regular content and help establish your company as an expert in your field. Promoting an e-book is a great way to build your brand's authority.

Book reviews:

Book reviews let you share your thoughts on books you've read about a specific topic. Be sure not to spoil any surprises in the story. You can also ask your readers about their favorite books to boost engagement.

Product reviews:

Product reviews are a great perk of having a successful blog or website, as companies often send you products for free to review. You can also buy products to review, avoiding any ethical issues that come with accepting free gifts.

Recipes:

Who doesn't like cooking? Just make sure that you only post original recipes. While ingredient lists aren't subject to copyright restrictions, the instructions on how to cook a dish are. So, ensure your cooking instructions are your own to avoid legal issues. Sharing original recipes adds uniqueness and authenticity to your content, making it more valuable to your audience.

Announcements:

Do you have a new employee, product, or even some new swag? Let your readers know about it in an announcement post. Market your site as the place readers can learn about news from your company before any other source.

FAQs

Everyone has questions. A FAQ (frequently asked questions) page is one of the best content types to help address common questions all in one place. This helps you better manage your time and not have to answer the same question over and over again

6. Explain the various types of content formats in WCMS. (4/6)

7. Explain various text file formats.(4/6)

8. Explain various graphics file formats.(4/6)

9. Explain various audio file formats.(4/6)

=CONTENT FORMATS:

These are the ways you present your content to your audience, based on its medium, style, or layout. For example, you might have content formats such as text, video, audio, image, infographic, or carousel.

Text File Formats:

Plain text files usually have the extension '.txt'. They are also called ASCII text files and can be viewed with an editor (such as Edit or Notepad) or with a Word Processor (such as MS Word or Word Perfect). ASCII stands for American Standard Code for Information Interchange. It was used for character representation in computers. However, the ASCII has now been replaced by Unicode. The characteristic feature of plain '.txt' files is that they do not allow any kind of formatting on the document (such as bold, italics, font colour, images, etc).

❖ .htm/.html/ files

These files are also text files that deserve a special mention as they are the 'language' in which web pages are authored. 'html' stands for Hyper Text Markup Language. The code of a web page is written in plain text and is saved with the extension '.htm/.html'. The web browsers (such as Mozilla Firefox, Internet Explorer, Google Chrome, etc.) identify the file as a web page, read the code and display it on the screen as we see it with the images, colours and hyperlinks.

❖ .xml files

XML, or eXtensible Markup Language, is a text-based format derived from SGML (Standard Generalized Markup Language). It's used for storing and exchanging data. XML is versatile and commonly used for structuring information in various web products and services, facilitating the exchange of formatted documents between different systems.

❖ .doc files

A very common format found on PCs, for formatted text files, '.doc' stands for document files. These files may be created, viewed and edited using programs such as MS Word. Several formatting features such as bold, italics, justification, bulleting, etc. are possible. It is a proprietary format of Microsoft.

❖ .odt files

ODT stands for Open Document Text. It is part of the Open Document Format (ODF) originally developed by Sun Microsystems Inc., but now maintained by Organization for the Advancement of Structured Information Standards (OASIS). It is an XML based format. The files use '.odt' file extension. They are originally supported by Open Office and the Star Office word processor. ODT format includes several XML files and configuration files in zipped format. A zip file is a collection of many files. These files can be viewed with any archive managing software, like Winzip.

❖ .pdf files

'PDF' stands for Portable Document Format. This file format was developed by Adobe Systems in order to make it possible to transfer formatted documents over the Internet so that their appearance would not change on any system. The biggest advantage of .pdf files is that it allows for printing of web pages – page by page as though it is a document file. This file type requires pdf reader for viewing e.g. Adobe Acrobat, XPDF, etc. This software can be downloaded from the Internet. .ps files 'PS' stands for 'Post Script' files. It is also an ASCII file type that is technically plain text. However, it is unreadable unless an on-screen viewer like 'Ghostscript' or any other postscript viewer is used.

Graphic Files:

Graphic file formats are many in number. Images are most important feature of web pages or any kind of publishing. It adds value as well as attracts the readers. In the Web parlance, images have to be capable of downloading quickly, they should not be bulky, though the original resolution should be preserved.

There are several Graphic file formats. Graphic styles may be divided into two major types:

- Raster Graphics
- Vector Graphics

Raster Graphics: Raster Graphics/Images are collection of dots or pixels. They are also called as bitmaps. The colour of each pixel is described by one or more information 33 Collaborative Content Development channels – separated into the primary hues – Red, Green and Blue or in a Single stream of colour mapped data. Raster images are simple images and are hence most suitable for interoperability. However, the primary disadvantage is that they do not scale well. Scaling may lead to a loss of resolution and hence poorer picture quality.

Vector Graphics: The more complicated of the two is Vector Graphics. They define an image as a collection of vector equations. The advantage of vector graphics is that it gives smooth curves and lines irrespective of the size of the image or resolution. However, the disadvantage is that they take longer to draw and require more storage space.

❖ .bmp files :

Bitmap files or .bmp files are the standard Windows Raster format. These files lay emphasis on quick display. They store images in the uncompressed form. The obvious trade off is that bmp files occupy lot of space. These files are quickly downloadable but on each transfer over network the quality of picture is lost.

❖ .cgm files

Computer Graphics Metafile (CGM) is an ANSI standard graphic file format for 2D vector graphics and raster graphics. CGM is a metafile that is a file containing information that describes or specifies another file. Therefore, it holds data and information for reconstructing graphical images. It was originally designed and used for clip art libraries, but is now mainly used for technical applications such as CAD drawings.

❖ .gif files

One of the most popular graphic file formats on the Internet, Graphic Interchange Files (.gif) was developed by CompuServe with the main purpose of archiving information. The .gif images are usually scanned stand-alone pictures that are not 'drawn' using an application Program.

❖ .jpeg/.jpg files

JPEG stands for Joint Photographic Experts Group that designed this format for high compression. It is one of the most popular image formats on the Web. It discards extra data and hence has good compression capabilities. It is a web standard and stores images in small size. It is useful when large numbers of files are to be incorporated and support millions of colours. It has better compression in photographs as compared to .gif

❖ .JAS files

JAS format is from JASC Inc. This file format is designed to create the smallest possible image file for 24 bits per pixel colour images and 8 bits per pixel gray scale images. Saving and retrieving an image using the JAS file format will result in some loss of image data. It has very high level of compression.

❖ PICT files

This is the standard Apple Macintosh graphic file format. It is accepted by many applications and imported/exported using clipboard (cut, copy, paste) to almost any text or graphics program. Raw files This flexible format consists of a stream of bytes that describes the color information in the file. Each pixel is described in binary format where 0=black and 255=white. This format is used to transfer documents between different applications.

❖ .tiff files

TIFF stands for Tagged Image File Format. This format was designed to overcome the problem of application dependence. It was originally designed to become the standard format. The format was intended to be capable of handling just about any possibility. This file format is generally used when graphic files need to be moved between different computer types (For example: PC to Mac and viceversa). These files allow for high resolution and are supported by most scanning and image editing software. This format works well for both on-screen display and print photographs. This format differentiates among three categories of images i.e. black and white, grayscaled, and colour.

Audio File Formats:

Sound files or audio files are gaining popularity on the Web. Today, most of the latest soundtracks are available on the Internet as sound files. There are even a few albums that have their presence only on the Web. Another popular application is online live news broadcasting. There are audio enabled websites. The following section discusses some of the common audio file formats.

❖ .au files

It is a simple audio file format which is most commonly found on the Web. It is required by PC users to load applications such as Waveform Hold and Modify (WHAM) to play these files. Macs need different sound applications to play this file type.

❖ .mid files

The .mid file extension is the standard extension used by MIDI music files. MIDI is a technical standard for Musical Instrument Digital Interface. These are used mostly in audio control in the multimedia industry. The MID files are used by a variety of music authoring programs and MIDI hardware devices

❖ .MIDI file specification allows for lengths to be specified as a variable number of bytes.

❖ .aiff files

Audio Interchange File Format (aiff) was developed by Apple. Although it was originally made for Macs, now it can be used by other platforms too. It is a good audio file format for use on the Internet. It can also be used in multimedia authoring on both Macs and Windows.

❖ .mp3

MP3 stands for MPEG layer three. It is currently the most popular audio file format. Its hallmark is its CD quality of music. MP3 allows for very high levels 35 Collaborative Content Development of compression. A minute of music may constitute approximately 1 Mb file. An MP3 player, is readily available on both Macs and Windows, is required to play this file type.

❖ .voc files

Creative Lab's Sound Blaster uses the .VOC file. They are designed for storing digitized voice data and hence the name. They can also handle any digitised sound in any format. The VOC files have a two part structure. The header block which defines the contents of the file and the data block which actually contains the audio information.

❖ .wav

Wav file is a commonly used file format on Windows machines. It can be used on the Internet and is good for multimedia authoring. It is flexible and handles both compressed and uncompressed storage formats. Video Files Formats Video files have become most popular with films being available and viewed on VCDs and DVDs. However within a multimedia lab, it is important to be aware of the video file formats as these are the most disk space-occupying (bulky) types.

❖ .avi files

Audio-Video Interleave (AVI) file format was developed by Microsoft. It is called 'Interleave' because the video and audio are bound together in chunks. The AVI file format is a very popular video file format and many videos have been produced in the format. AVI files can be opened in most of the video players such as VLC (Video LAN Client), RealPlayer, and Microsoft's Windows Media Player.

❖ .mov/.movie files

Movie files are the common format used in QuickTime movies. It is a multimedia format developed by Apple. Originally, it was designed for Apple machines but now it is widely used in websites for streaming audio or video.

❖ .mpg/.mpeg files

This is another standard format. This format uses MPEG compression scheme of audio and visual (AV) digital data. MPEG video is a series of video standards defined by the Moving Picture Experts Group (MPEG) External Link.

10. Write a note on content tools. (4/6)

=Content Tools (Media-wise) :

Content creation involves human intellect, but tools enhance efficiency in expressing ideas. Text, graphics, audio, and video are common mediums, each requiring specific tools for creation. These tools empower creators to effectively convey their thoughts and ideas to their audience in diverse and engaging formats.

Based on the form of media, tools can be categorized as:

- Text editing tools
- Graphic editing tools
- Audio capture and editing tools
- Video capture and editing tools

Text Editing Tools: Text editors are software used in Desktop Publishing (DTP). These are editors such as MS-Office Suite, Open Office Suite and Star Office, etc. They include packages to type text and also proof read. Other than this, they also contain packages for presentation, spread sheets and database for powerful presentation, calculation, and storage and retrieval respectively. A lot of literature already exists which requires retro-conversion from print to digital. Optical Character Recognition (OCR) technology is used to convert printed text to digital. Software like Omni Page Pro, Text Bridge, Abbyy FineReader, etc. are OCR software available in the market. However, nowadays scanners come with basic OCR utility which can be used for light weight text editing.

Graphic Editing Tools : DTP is supplemented by effective use of graphics (images). This includes capturing the image and processing it in a presentable format. This requires use of capture devices like camera (ordinary as well as digital camera), scanners and data transfer cables. Once an image is captured it is to be converted to usable format and size. However,

sometimes value addition to the image is also done by touching up the image. For the purpose, software like Adobe Photoshop, Corel Draw, etc. are widely used by professionals. However, GIMP (GNU Image Manipulation Program) is also a powerful tool to edit graphics available under Open- Source License.

Audio Capture and Editing Tools: Sound has more pneumatic property than text and graphics. Hence, libraries are going more and more towards recording and collecting audio materials. Audio can be recorded with a recorder which is inbuilt in Mobile phones, Dictaphones, Computers, etc. These are affordable devices used commonly in day-to-day affairs. However, there are very costly studios with various kinds of mixers and error correction devices which are also available and are used in professional recording and music. Audio Capture Card is available in the present day computers which can be used for recording. The card is connected with a Microphone which records input and feeds to the card. There are inbuilt utility software in the operating system which can be used to record and save the voice. Professional software like Authorware, WavePad, Goldwave, etc. can be used to create mixing effect.

Video Capture and Editing Tools: Libraries have lot of recorded presentations in their collections. The recording of any event is captured with the help of a video camera. Nowadays video camera directly record in computer readable format, therefore, there is no need to attach Video Capture Card with the computer. For editing the video there are many options available for example, CyberLink PowerDirector, Corel Video Studio ProX4, Adobe Premiere Elements, MAGIX Movie Edit Pro, VideoPad, Pinnacle 31 Collaborative Content Development Studio HD, Roxio Creator, Sony Vegas Movie Studio Platinum, Roxio Video Lab, ShowBiz DVD and many such others.

11. How to create digital graphics? (6)

=Most graphic designs today are digital, created on computers. The elements of the graphic design are arranged in an attractive manner using specialized graphics software. Once the digital graphic design is complete, it may be printed with a computer printer, or sent to a commercial printer.

Things You'll Need:

- Computer
- Graphics software
- Pencil and paper
- Printer
- Photo or illustration, optional
- Scanner, optional

Step 1

Prepare your design idea with pencil and paper. Decide what words, images, colors and shapes to use in your digital graphic design. Use the pencil to make a few rough sketches of how your design should look. Try several different layouts, combining the typography, images and shapes in an attractive manner that effectively communicates your message. Color information may be

noted to the side of the rough sketch, or added using color pencils. These are quick sketches to develop your design idea, not finished designs.

Step 2

Turn on your computer and open a file in a graphics program. Among the more popular graphic design programs are InDesign, Photoshop and Illustrator from the Adobe software company, and Corel Graphics Suite. Open Office Draw may be downloaded for free (see Resources).

Step 3

Name the file and save it. Type in the words you will use in your design. Select a font for the words. Resize the words and place them according to your rough pencil sketch. Select a new color for the words if indicated in your rough sketch. Your words have become typography for the purpose of your digital graphic design.

Step 4

Import any existing photo or illustration files you need into your digital graphics file using the design software's "Import File" feature. If you only have a hard copy of the photo, use a scanner to create a digital file. After importing, resize as needed and position in your design as per your rough pencil sketch. You may also simply open the photo or illustration file, copy it and paste it directly into your digital graphics file.

Step 5

Use the tab features in the design software to add graphic and color elements as per your rough pencil sketch. Borders may be added to photos, for example. A variety of shapes may be added and colorized as needed.

Step 6

Print out a copy of your digital graphic design to check it. Make changes as desired. Print your final copy or copies, or copy your project onto a CD or memory card and have it commercially printed. In some cases, you may be able to email the project directly to a commercial printer.

12. Explain needs and general guidelines for content development. (4)

=Needs:

In the vast world of online information, both users and hosts face challenges. Users struggle to find relevant information due to its sheer volume and inconsistent presentation. Hosts aim to attract target audiences but face difficulties in organizing content effectively. Standards and guidelines have emerged to address these issues, focusing on structuring documents and presentation. Content creation for the web requires adapting to user behavior, with concise, easily scannable content. Hosts must also consider content quality, ensuring it's trustworthy and safe. Standardization is crucial for global accessibility and interoperability, enabling easy reuse of information across platforms and languages.

Some General Guidelines :

When using the internet and electronic documents, people often face problems like files not working together, slow downloads, and hard-to-understand content. To make sure users get good quality, content creators should follow some basic rules.

According to Jacob Neilson, a web content analyst, the content developers should take care of the following points:

- i) Well-tested and widely used packages in designing web pages should be used. It is better to avoid recently released software, especially if there is use of plug-ins such as Flash or PDF. A good rule is not to use upgraded versions immediately till they mature. Software vendors require time to fix their bugs and offer reasonably reliable versions. It also gives users time to upgrade.
- ii) As far as possible Scripts in web pages should be avoided. Adding code means that there is also a risk of bugs (or faults). If scripts are a must, then they should be tested and debugged on all combinations of platforms and browser versions. Norms and Guidelines for Content Development 333 Content Development.
- iii) It is advisable to have simple point-and-click web navigation. Special menu controls often fail, especially for users who move their mouse fast or who have motor skill impairments.
- iv) The website should always be made as compact in size as possible. It is good to have spare server capacity so that your site can cope with traffic surges.
- v) Server software must be robust. When selecting vendors, ensuring code quality must be a priority.

13. Explain the content development guidelines according to Mick Wood. (6)

=According to Mick Wood, following guidelines help content developers to cater for the needs of all Internet users.

Screen Layout Screen layout should be consistent. The more consistent a website is in its design, the easier it will be for users to navigate. Users, especially persistent users, tend to learn and remember the location of key, functions and controls.

Logos Screen layout should include logos, navigation buttons and footer information. Putting the logo in a consistent place on every page (usually top left) ensures that visitors are fully aware that they are on the same site.

Update Information 'Page Last Updated' information needs to be included. Visitors will quickly know if you have added any new content to your site, and it may prompt you to do so. The date format should be understandable, i.e., 1st February 2003 is better than 02-01-2003. In the UK 02-01-03 means 2nd January 2003 and in USA it means 1st February 2003.

Links A link to the home page should be added to every page except the homepage itself. Pages should not link to themselves as this can confuse some users. Making your logo the home page link is considered to be a good practice.

Image Sizes Considerable number of Internet users still use modems with connection speeds of 33K or less. Pages should therefore be kept to a maximum of 35K ensuring download times of less than ten seconds for these users. Visitors may not wait if your pages take too long to download. All images, including spacer images, should include alternative text. The alternative text should describe the function of the image, if applicable:

Use of Frames should not be used unless absolutely necessary. Frame-based sites can be confusing for the visually disabled, particularly those using -screen readers or speech browsers - users can easily become disorientated, Additionally, users cannot easily bookmark individual pages within a frame-based site.

Page Title Each page should have a descriptive and different title, clear and helpful headings, and a logical structure. Titles are used by search engines to identify web 'pages, If two or more pages have the same title they cannot easily be differentiated by users.

Font Attributes The FONT tag should generally be avoided in creating accessible websites. While attribute specifications like SIZE="+ 1" or SIZE="-1" are relatively harmless, absolute sizes like SIZE="1" can result in text that is too small to read. Instead: Cascading Style Sheets should be used to format web pages. Style sheets allow authors to suggest relative changes in font size with much greater flexibility than is allowed under FONT.

Color Attributes The COLOR attribute of the FONT tag should always be avoided since many browsers still display the font color when the user tries to override author specified colors. The result could be an unreadable document if the font color does not contrast well against the reader's chosen background.

Text Size and Font Text should be the equivalent of size 12 points to enhance reading performance. Research has shown that there is no noticeable difference in reading speed or user preference between Times New Roman, Georgia or other serif fonts and Helvetica, Arial or other sans-serif fonts. The FACE attribute of the FONT tag cannot be overridden by the user in many browsers. This may result in the author choosing a font that is very difficult to read given the user's platform and environment settings. The same font may display strangely on different platforms.

Link Identification Links should be clearly identified and it should make sense when reading out of context. Links should never be designated with the text 'click here'. Some screen readers can be set to read out a list of links on a particular page; a list of 'click here' links is not helpful. . Blue underlined text is the preferred choice for all links. Some users miss links because the text is not underlined. Research shows that users can easily find links which include visual cues, that

is, links that are underlined, rather than having to move the mouse to see when the pointer changes to a hand (this is known as mine sweeping).

Visited links should be designated using a different colour. Many users use link colours to identify which parts of a site they have already visited.

Page Display. Pages should be fluid flexible because users browse the Internet using a variety of screen resolutions. Pages should be designed to expand or contract according to the user's settings, thus preventing horizontal scrolling. Horizontal scrolling is a particular problem for screen reader users because the screen reader does not automatically scroll horizontally - users may miss important content.

Device Independence Pages should be device independent. Not all users navigate websites using a mouse - many disabled users use Assistive Technology. All users should be able to navigate the site using the input device of their choice. Keyboard shortcuts, using the 'access key' attribute, should be provided for key links.

Readability Paragraphs and sentences should be kept short. Readability improves when sentences and paragraphs are kept relatively short. Users tend to scan web pages and will often skip over large chunks of text.

While these guidelines aren't formal, hosts of electronic documents should still adhere to them. Content and organization issues are being looked at closely to establish standard norms and guidelines for content development. Organizations like the International Organization for Standardization (ISO) are involved in this process. For instance, ISO 8879 deals with SGML, the Standard Markup Language for the web, and has been adopted by initiatives like the Text Encoding and Interchange (TEI) for structuring web documents.

14. Explain steps involved in the audio editing process.(4)

=There are four main steps involved in the basics of audio editing:

- Selection
- Editing
- Processing
- Output

Selection is the first step in audio editing. This is where you identify the portion of the audio that you want to edit, and it can be done manually or with the help of software tools.

Next is **Editing**, where you make changes to the selected audio. Again, many different techniques can be used here, and the exact method will depend on the software being used.

Processing is the third step, where you add effects or make other changes to the processed audio. Several different plugins or software programs can be used for this. Most audio processing programs will offer a suite of tools to help you get the sound you need. These can

include EQ, compression, limiting, reverb, and delay. There are many different ways to process audio, so it's important to find a program that offers the specific tools required for your project.

Output, where you save the edited audio file. The most common audio formats are WAV and MP3, but there are many others. WAV files are uncompressed and will take up more space than MP3 files. They are best suited for applications where sound quality is paramount, such as music editing or DVD authoring. MP3 files are compressed and will take up less space than WAV files. They are ideal for applications where file size is crucial, such as streaming audio or portable devices.